

6. Further matrix algebra

What students need to learn:

Linear transformations of column vectors in two and three dimensions and their matrix representation.	Extension of work from F1 to 3 dimensions.
Combination of transformations. Products of matrices.	The transformation represented by $\mathbf{AB}$ is the transformation represented by $\mathbf{B}$ followed by the transformation represented by $\mathbf{A}$.
Transpose of a matrix.	Use of the relation $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.
Evaluation of 3×3 determinants.	Singular and non-singular matrices.
Inverse of 3×3 matrices.	Use of the relation $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$.
The inverse (when it exists) of a given transformation or combination of transformations.	
Eigenvalues and eigenvectors of 2×2 and 3×3 matrices.	Normalised vectors may be required.
Reduction of symmetric matrices to diagonal form.	Students should be able to find an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is diagonal.
